ASHUTOSH MOHAPATRA

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RESEARCH INTERESTS

Cloud & distributed systems optimization, ML/RL for adaptive resource management, Energy-efficient sustainable computing and Geospatial big data analytics.

EDUCATION

ODISHA UNIVERSITY OF TECHNOLOGY AND RESEARCH — BHUBANESWAR, IN
B.Tech, Computer Science & Engineering — May 2027 (Expected)
CGPA. 9.4/10

BHUBANANANDA ODISHA SCHOOL OF ENGINEERING — CUTTACK, IN
Diploma, Computer Science & Engineering — May 2024
Grade. 89.3%

RESEARCH EXPERIENCE

UNIVERSITY OF TORONTO — TORONTO, CA
Research Intern, Department of Computer Science — May 2026 – Present
(Supervised by Prof. Eyal de Lara & Mr. Pritish Mishra)

Investigating structural constraints in distributed vector database architectures, focusing on the engineering of high-throughput range filtering mechanics and indexing optimizations for real-time streaming spatiotemporal data.

KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY (KIIT) — BHUBANESWAR, IN
Research Collaborator, School of Computer Applications
(Supervised by Dr. R. K. Barik) — *April 2026 – Present*

Synthesizing multi-tier optimization policies at the intersection of green cloud architectures and geospatial big data systems, exploring the deployment of quantum-accelerated spatial processing pipelines.

ODISHA UNIVERSITY OF TECHNOLOGY AND RESEARCH — BHUBANESWAR, IN
Undergraduate Research, School of Computer Sciences — June 2025 – Present
(Supervised by Dr. Meenakshi Pant)

Architected energy-aware cloud system models using simulation traces to optimize framework scalability, while currently investigating quantum-classical hybrid scheduling frameworks within dedicated cloud computing laboratories.

PUBLICATIONS

AL-WOA: ADAPTIVE LEARNING WHALE OPTIMIZATION ALGORITHM FOR ENERGY-AWARE LOAD BALANCING IN CLOUD DATA CENTERS
Expert Systems with Application — [DOI]

Designed a hybrid reinforcement learning–assisted metaheuristic optimization framework for cloud load balancing, achieving improved convergence speed, reduced energy consumption, and enhanced SLA compliance across real-world Google and Alibaba cluster workloads.

RESEARCH PROJECTS

PREDICTIVE QUANTUM RESOURCE SELECTION FOR HYBRID CLOUD COMPUTING USING ADAPTIVE DECISION INTELLIGENCE
Submitted

Developed an Adaptive Decision Intelligence framework using gradient-boosted learning to route workloads between classical and quantum resources, cutting average job latency, cost, and quantum processor time while achieving 93.5% routing accuracy against quantum-first and classical-first baselines.

A Cloud-Native Geospatial Big Data Digital Twin for Real-Time Urban Flood Risk Decision Support
submitted

Built a cloud-native digital twin fusing open geospatial layers with simulated real-time sensor streams for near-real-time flood risk scoring, validated on Chennai's Velachery flood zone with sub-second response for 275+ users and 75% agreement with GIS baselines.

Cloud-Integrated Digital Twin Framework for Intelligent Monitoring and Maintenance of Civil Infrastructure
Under peer review

Designed a cloud-based digital twin integrating real-time sensor synchronization, ML-driven damage prediction, and metaheuristic optimization for civil infrastructure, improving alert precision and reducing cumulative maintenance costs over conventional and partially integrated monitoring baselines.

A Cloud–AI Digital Twin Framework for Decision-Intelligent Resilience of Marine Infrastructure
Revision submitted

Proposed a cloud-hosted AI digital twin combining sensor fusion and decision intelligence for proactive marine infrastructure resilience, improving alert precision and recall, reducing response latency under load, and cutting maintenance costs versus cloud-only and ML-based baselines.

Hedgehog Hibernation Optimization (HHO): An Energy-Efficient VM Migration Algorithm for Green Cloud Computing
Under peer review

Developed a biologically inspired VM consolidation algorithm integrating adaptive migration and server hibernation strategies, resulting in measurable reductions in energy consumption, SLA violations, and migration overhead compared to ACO, PSO, and MBFD baselines.

An Optimized Hybrid Cloud Framework for Hadoop-Driven Big Data Visualization in Power BI
Revision submitted

Designed and evaluated a hybrid cloud analytics architecture integrating Hadoop, Azure Data Lake Storage Gen2, and Azure Synapse Analytics, enabling low-latency Power BI DirectQuery access with improved ingestion efficiency, query performance, and dashboard interactivity on large-scale datasets.

EMPLOYMENT

National Thermal Power Corporation — Talcher, Odisha, IN
IT Support Engineer (Intern), O&M Dept. — June 2025 – July 2025

Gained hands-on exposure to critical IT infrastructure, SCADA operations, and industrial cybersecurity workflows within a large-scale energy environment to understand technology-driven operational efficiency.

Tata Consultancy Services — Bhubaneswar, Odisha, IN
Jr. Web Developer, R&D Dept. — July 2023 – December 2023

Developed responsive user interfaces and supported front-end integration using modern web technologies within an agile framework to enhance platform usability and performance.

REFERENCES

Dr. Subasish Mohapatra
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Odisha University of Technology and Research, Bhubaneswar, India
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Dr. Sachi Nandan Mohanty
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